

MT 201: Physics of Solids

Course Code	MT 201
Course Name	Physics of Solids
L-T-P-C	3-0-0-3
Prerequisites	None
Intended for	B.Tech. (MSE)
Mutual Exclusion	None
Approval	57th BoA (Oct 2024)

Course Contents

- **Crystal Structures:** Elementary cell (Translation vectors, Primitive translation vectors, Bravais lattice, Basis, Crystal systems, Unit cell, and Bravais lattice types), Symmetry transformations (Axis of symmetry, Mirror planes, Point and Space symmetry groups), (Reciprocal space, Reciprocal lattice. Bragg planes, Brillouin zone, Ewald sphere Surface and Interface Physics (surface crystallography), Bragg conditions (Methods of structural studies: Laue method, rotating crystal method, powder method), Ewald construction, Lattice defects, Burger's vector. [12 Hours]
- **Transport Properties:** Free Electron Theory, Fermi-Dirac distribution function, and Density of states (DoS). Energy bands in solids, electronic band structure of solids Bloch theorem and energy band, Effective mass theory, Boltzmann transport equation, Thermal and electrical conductivity (for metals, dielectrics, semiconductors), Lattice dynamics, Electron and phonon scattering, Thermal properties, the specific heat of solids, 2D electron gas, Quantum wells & semiconductor superlattices. [10 Hours]
- **Optical Properties:** Drude theory-free carrier contribution to the optical properties, Absorption of light in solids, Optical properties of solids over a wide frequency range, Impurities and excitons, Luminescence and photoconductivity, Optical study of lattice vibrations, Nonlinear optics, and Electron spectroscopy and surface science. [6 Hours]
- **Magnetic Properties:** Hund's law, Spin-orbit interactions, Heisenberg exchange interaction, Diamagnetism and paramagnetism, Ferromagnetism and antiferromagnetism, Curie & Neel point, Landau Diamagnetism, Magneto-resistance, Nuclear magnetic resonance (NMR), and Kondo effect. [5 Hours]
- **Superconducting Properties of Solids:** Review of superconducting properties of solids, superconductivity, Meissner effect, London equation, Type I and Type II superconductors, High temperature superconducting materials, Electron orbitals, Macroscopic quantum description of the supercurrent. [8 Hours]

Text and Reference Books

1. H. Ibach and H. Luth, **Solid-State Physics, An Introduction to Principles of Materials Science**, 4th Edition, Springer, 2009.
2. C. Kittel, **Introduction to Solid State Physics**, 8th Edition, John Wiley & Sons, 2005.
3. N. W. Ashcroft and N. D. Mermin, **Solid State Physics**, Brooks Cole, 1976.
4. David W. Snoke, **Solid State Physics: Essential Concepts**.